SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	**Assessment:**	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO3 Statement:	*Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign.(BL5)*		
Student Name:	Nivya Sri T	**Reg. No.:**	192424131
Section / Batch:	2024	**Date:**	18-08-2026

Code of Conduct

I _____Nivya Sri T________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _______Nivya Sri T_____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Visualization Critique & Redesign

CO3 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO3-AT3 Visualization Critique & Redesign problems, in which students are given a flawed or misleading chart, must critique its specific shortcomings, and propose a justified redesign.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the specific visualization principle being violated (e.g., proportional ink, axis scaling, color perception, overplotting) in the given flawed chart.	Good understanding of the relevant principle, with only minor gaps in identifying exactly what is violated.	Basic understanding, but with some misconceptions about which principle applies to the flaw shown.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	**25%**	Clearly explains why the flaw is misleading, including what a reader could wrongly conclude as a result — not just that something is 'wrong.'	Explains the flaw with adequate reasoning, covering most of why it misleads a reader.	Identifies the flaw but explanation is generic or only loosely tied to its misleading effect.	Little or no explanation of why the flaw is misleading; the critique is a bare assertion.
Application	**20%**	Proposes a specific, well-justified redesign that directly resolves the identified flaw, correctly applying appropriate chart-selection and design principles.	Proposes a workable redesign that resolves the flaw, with only minor gaps in justification.	Proposes a redesign, but it only partially resolves the flaw or lacks sufficient justification.	Fails to propose a workable redesign, or the redesign does not address the identified flaw.
Organization	**15%**	The response is clearly structured, addressing both the critique and the redesign in a logical sequence with coherent overall flow.	The response is organized and addresses both parts, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Accuracy	**10%**	Both the critique and the redesign are completely accurate and well-suited to the specific chart and data described.	Mostly accurate, with only minor omissions or a small error in the critique or redesign.	Partially correct; the critique or redesign contains notable inaccuracies or mismatches to the scenario.	Largely incorrect or inappropriate, with major gaps in both the critique and the redesign.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

PROBLEM 3:

a) Critique why a pie chart is fundamentally the wrong choice for this multi-select data.

A pie chart is fundamentally wrong because:

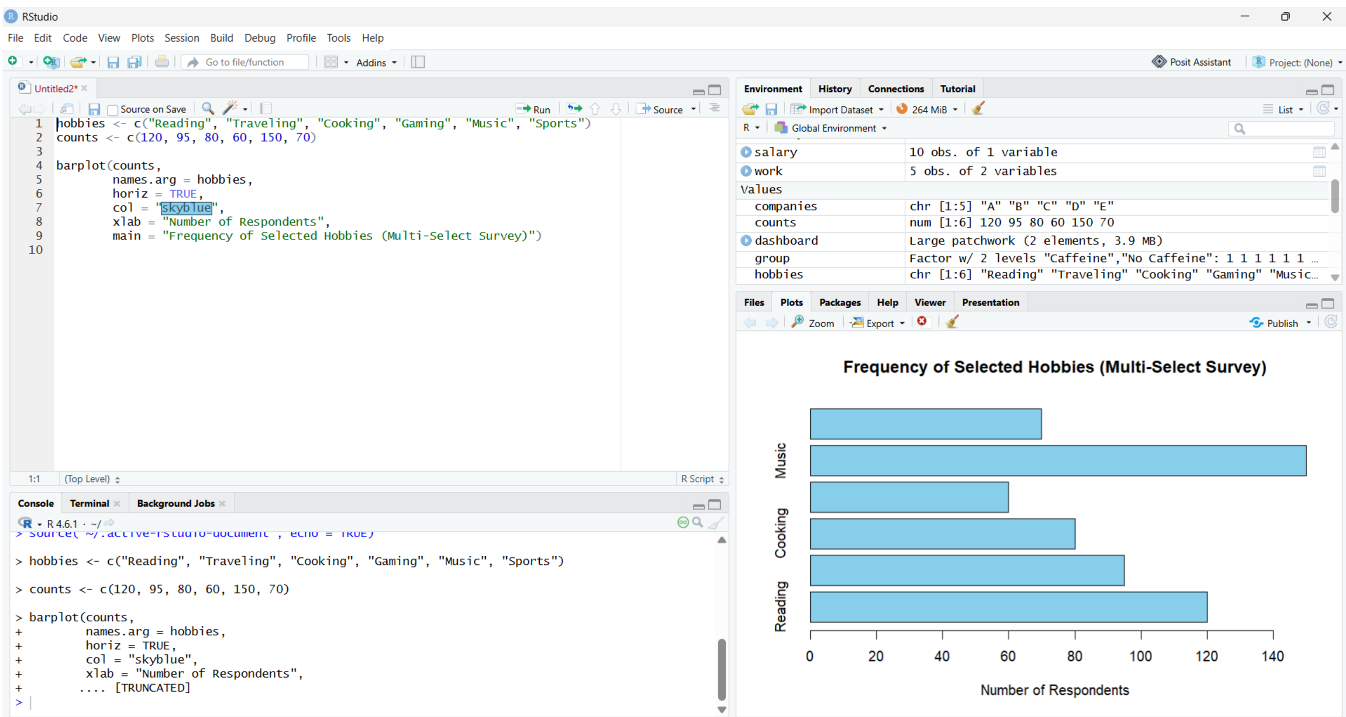
- It assumes mutually exclusive categories, but multi-select data allows respondents to choose multiple hobbies.
- It assumes all slices must sum to 100%, but multi-select selections naturally exceed 100%.
- It falsely implies relative share comparison, even though shares are meaningless when respondents select multiple options.
- It visually distorts the data by forcing overlapping selections into a single whole.

b) Redesign: propose an appropriate chart type for multi-select survey data.

Use a bar chart (horizontal or vertical) showing:

- Counts of respondents selecting each hobby, or
- Percentages (not summing to 100%).

This treats each hobby as an independent variable and avoids false “whole-part” interpretation.



PROBLEM 7:

a) Critique why percentage-only reporting like this can be misleading.

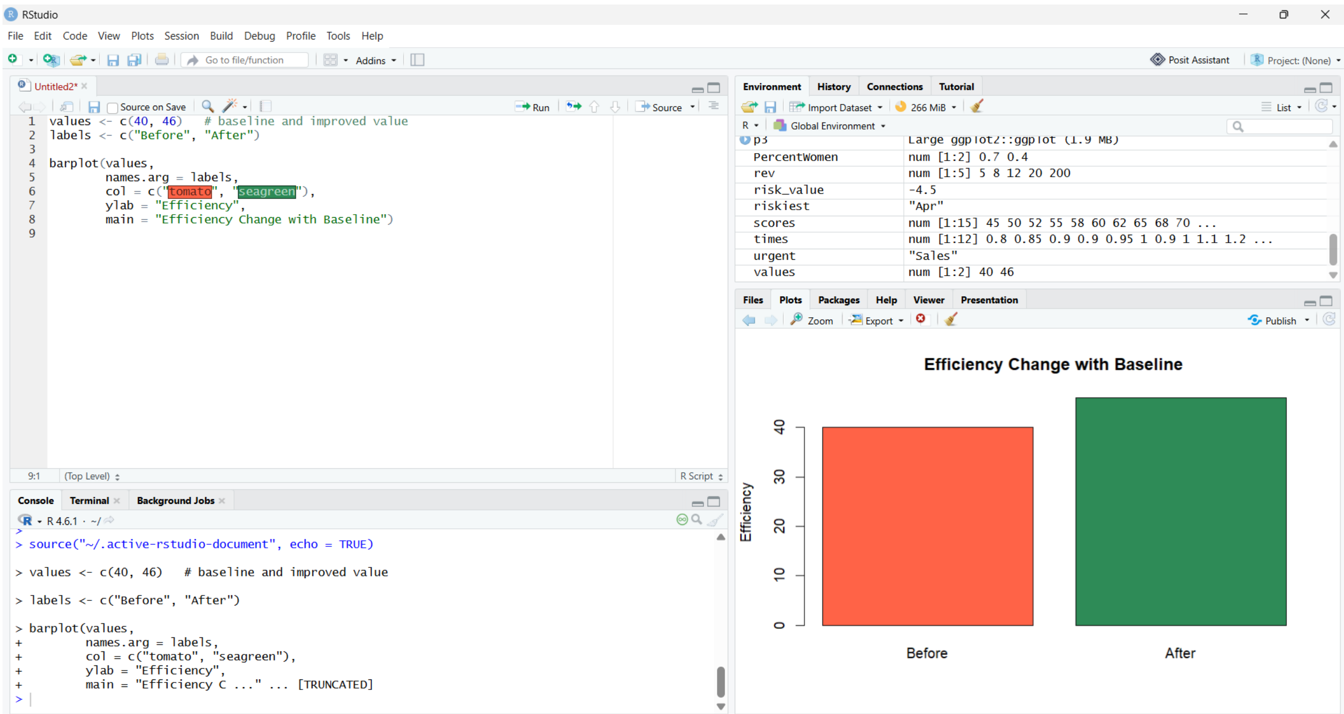
Percentage-only reporting is misleading because:

- A percentage change is meaningless without a baseline (15% of what?).
- It hides absolute values — an increase from 10 to 11 vs. 1000 to 115 are both “15%”.
- It prevents readers from judging real impact, scale, or significance.
- It can exaggerate improvement when the starting value is very small.

b) Redesign: what additional information or chart element would you add to fix this?

Add:

- Baseline value (e.g., “Efficiency increased from 40 to 46 units”).
- Absolute change (e.g., “+6 units”).
- A bar chart or line chart showing before vs. after values.



PROBLEM 11:

a) Critique the specific problem this creates for the reader.

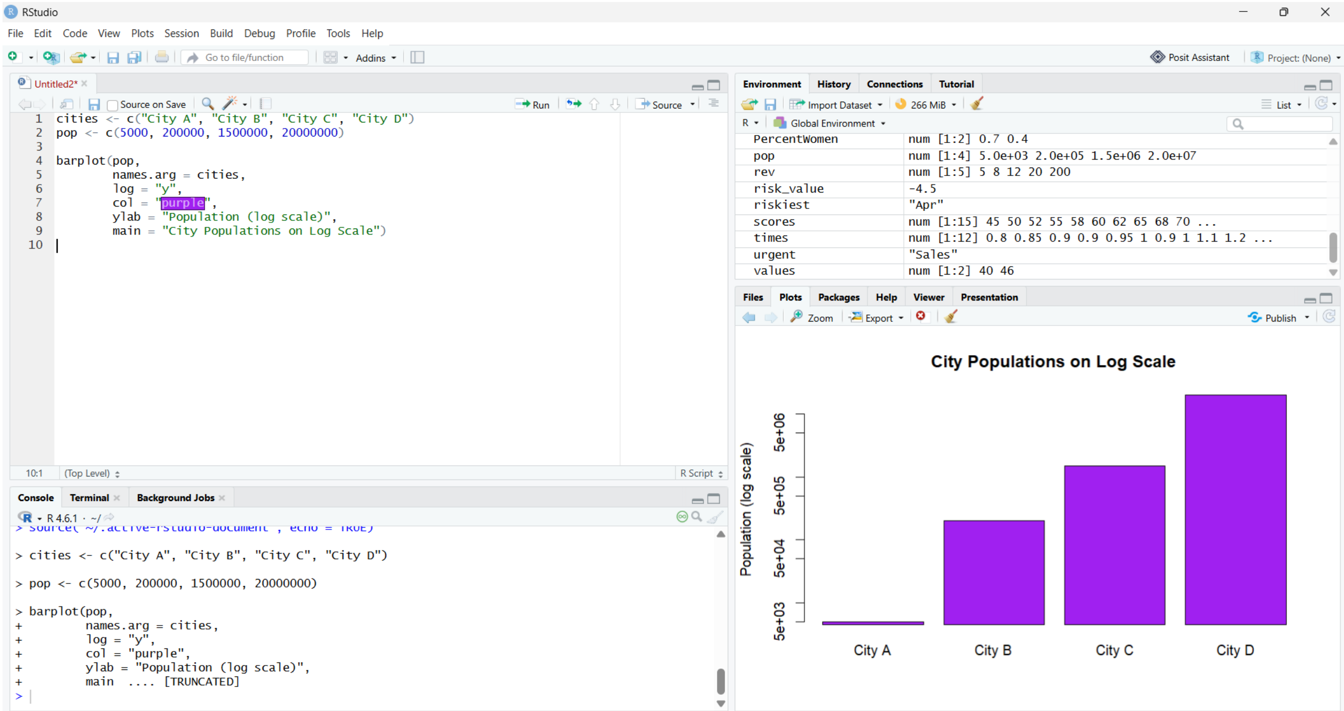
A linear y-axis creates:

- Severe scale compression — small populations appear as flat lines.
- Loss of comparative visibility among mid-sized cities.
- Misleading impression that only the largest city matters.
- Readers cannot perceive meaningful differences in smaller values.

b) Redesign: propose the correct axis type and justify it.

Use a logarithmic y-axis, because:

- Log scale handles large magnitude differences.
- It preserves relative differences across all cities.
- It makes small and large values visually comparable.



PROBLEM 20:

a) Critique the misleading impression created by drawing both circles the same size.

Drawing both pie charts at the same size is misleading because:

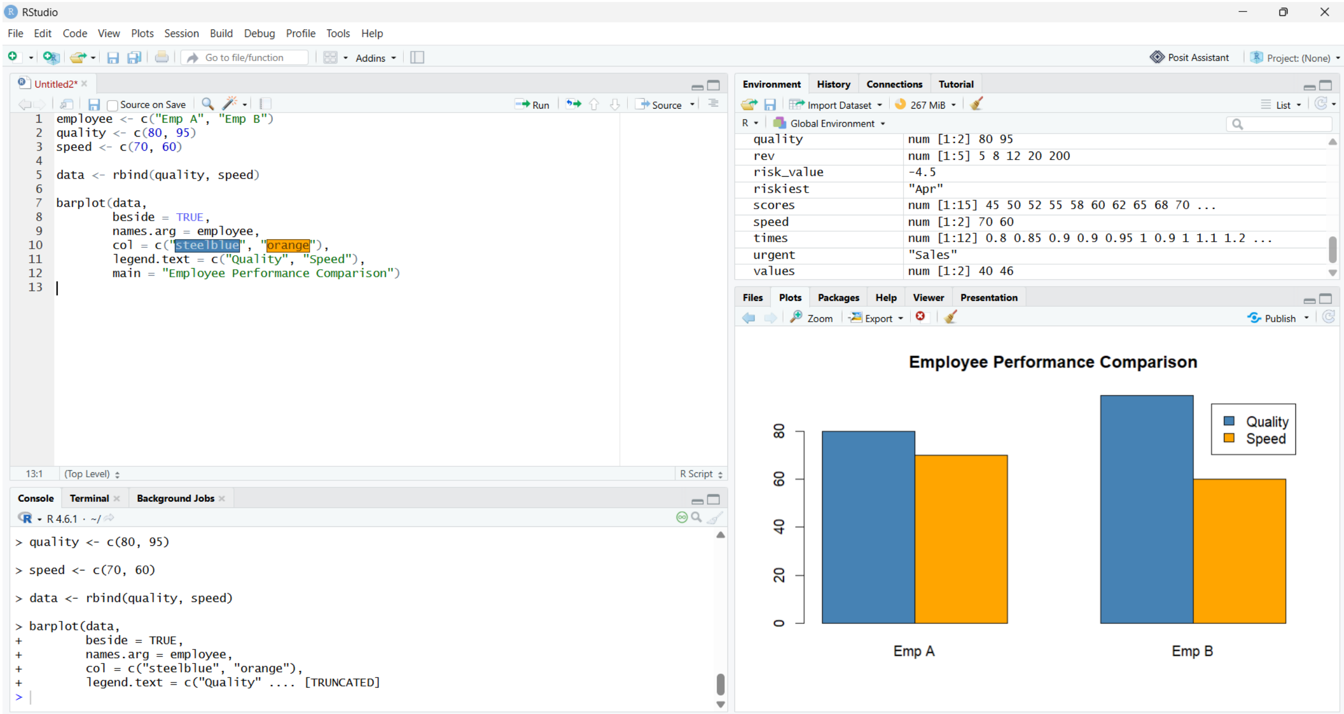
- It implies the employees have equal total performance, even if one has far more tasks.
- It hides differences in overall workload, total output, or scale.
- It forces comparison of percentages instead of actual performance.

b) Redesign: propose a fairer way to compare the two employees.

Use:

- A bar chart comparing absolute values (tasks completed, scores, etc.).
- Or a grouped bar chart showing categories side-by-side.
- Or a single stacked bar showing total + category breakdown.

This ensures fair comparison of both totals and components.



PROBLEM 24:

a) Critique the specific problem with using this many hierarchy levels in a sunburst chart.

Using 6 levels in a sunburst chart causes:

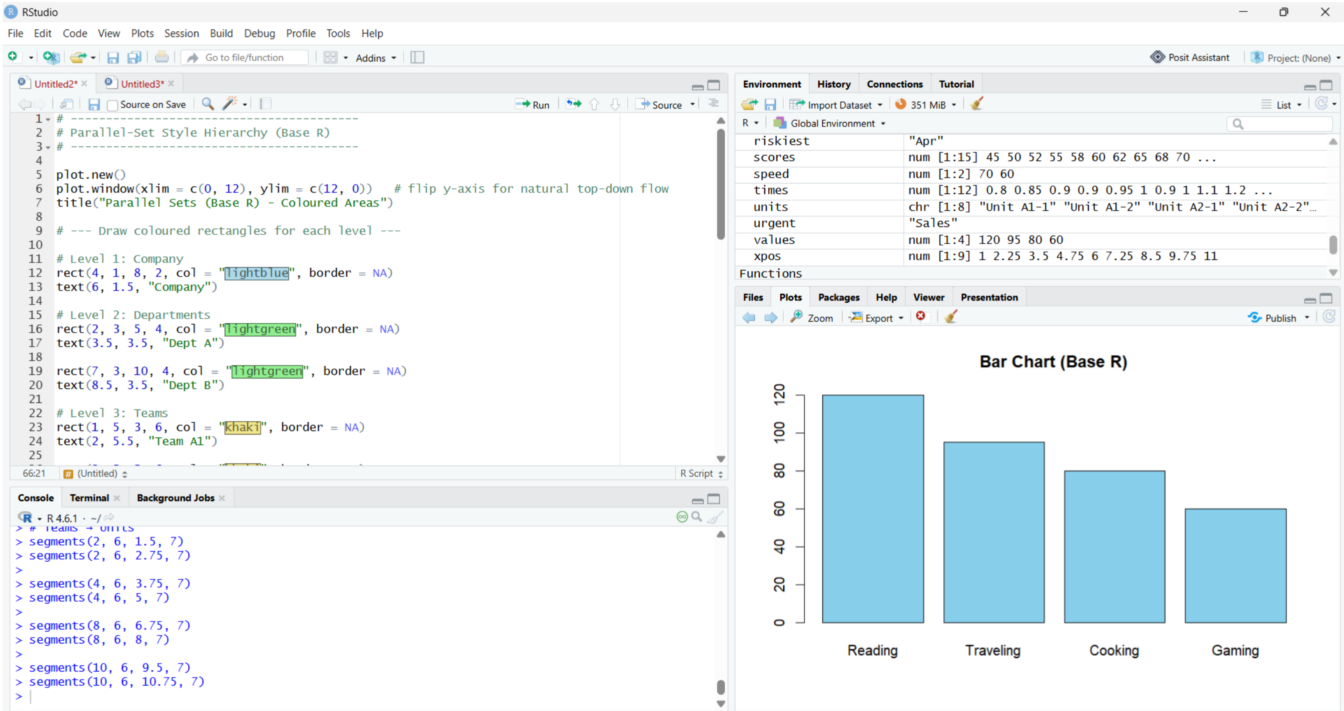
- Extreme radial compression — outer rings become too thin to read.
- Labels become overlapping or invisible.
- Users cannot interpret deep hierarchy relationships.
- Visual clutter overwhelms comprehension.

b) Redesign: propose an alternative approach for visualizing deep hierarchies.

Use:

- Indented tree diagram (best for deep hierarchies).
- Treemap (area-based, readable at many levels).
- Network diagram (for complex relationships).

These preserve readability and structural clarity.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
Untitled2*
Untitled3*
Source on Save
Run
Source
# --
# Parallel-Set Style Hierarchy (Base R)
# --
plot.new()
plot.window(xlim = c(0, 12), ylim = c(12, 0))   # flip y-axis for natural top-down flow
title("Parallel Sets (Base R) - Coloured Areas")
# --- Draw coloured rectangles for each level ---
# Level 1: Company
rect(4, 1, 8, 2, col = "lightblue", border = NA)
text(6, 1.5, "Company")
# Level 2: Departments
rect(2, 3, 5, 4, col = "lightgreen", border = NA)
text(3.5, 3.5, "Dept A")
rect(7, 3, 10, 4, col = "lightgreen", border = NA)
text(8.5, 3.5, "Dept B")
# Level 3: Teams
rect(1, 5, 3, 6, col = "khaki", border = NA)
text(2, 5.5, "Team A1")
66:21
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> segments(2, 6, 1.5, 7)
> segments(2, 6, 2.75, 7)
>
> segments(4, 6, 3.75, 7)
> segments(4, 6, 5, 7)
>
> segments(8, 6, 6.75, 7)
> segments(8, 6, 8, 7)
>
> segments(10, 6, 9.5, 7)
> segments(10, 6, 10.75, 7)
>
Environment
History
Connections
Tutorial
Import Dataset
351 MiB
List
Global Environment
riskiest "Apr"
scores num [1:15] 45 50 52 55 58 60 62 65 68 70 ...
speed num [1:2] 70 60
times num [1:12] 0.8 0.85 0.9 0.9 0.95 1 0.9 1 1.1 1.2 ...
units chr [1:8] "Unit A1-1" "Unit A1-2" "Unit A2-1" "Unit A2-2"…
urgent "Sales"
values num [1:4] 120 95 80 60
xpos num [1:9] 1 2.25 3.5 4.75 6 7.25 8.5 9.75 11
Functions
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Bar Chart (Base R)
0
20
40
60
80
100
120
Reading
Traveling
Cooking
Gaming